Supplementary material for "CertiVar: Prevalence-Free Certified Error Rates and Likelihood Ratios for Missense Variant-Effect Predictor Calls"

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Section, figure and table numbers without the prefix S refer to the main paper.

S1 Proofs

Assumptions, propositions, the theorem and the corollary are stated in Section 3.1 of the main paper; Q, C, $\hat{t}_k$, L^+ and U^- are defined there.

Throughout, fix a class k and write F_k for the distribution function of the nonconformity r_k of a variant drawn from $Q(\cdot \mid Y = k)$. By Assumption 1, the kept calibration variants (one per residue of $\mathcal{A}$) and the variant to be called are independent draws from Q; conditionally on all labels, the n_k kept variants of class k and a class-k test variant are independent draws from $Q(\cdot \mid Y = k)$.

Proposition 1. Conditionally on the labels, the $n_k + 1$ nonconformities of the class-k kept variants and of the test variant are exchangeable, so with ties broken at random the rank of the test value is uniform on $\{1, \ldots, n_k + 1\}$ [1, 2]. The test value is at most the j_k-th smallest calibration value with probability $j_k/(n_k + 1)$, which lies in $[1 - \alpha_k,\, 1 - \alpha_k + 1/(n_k + 1))$ for $j_k = \lceil (n_k + 1)(1 - \alpha_k) \rceil$; ties only increase the probability, which gives the lower bound in general. If $j_k > n_k$ then $\hat{t}_k = \infty$. Since $k \in C$ exactly when $r_k \le \hat{t}_k$, the claim follows. □

Proposition 2. Conditionally on the labels, the coverage of the threshold is $F_k(\hat{t}_k) = F_k(r^{(k)}_{(j)})$. If F_k is continuous, $F_k(r^{(k)}_{(j)})$ is the j-th order statistic of n_k independent uniforms and has the $\mathrm{Beta}(j, n_k - j + 1)$ distribution, so $\Pr\{F_k(\hat{t}_k) < 1 - \alpha_k\} = F_{\mathrm{Beta}(j, n_k - j + 1)}(1 - \alpha_k) \le \delta$ by the choice of j_k [3]. If F_k has atoms, $F_k(r^{(k)}_{(j)})$ is stochastically larger than in the continuous case, so the bound still holds. If no $j \le n_k$ qualifies, $\hat{t}_k = \infty$ and coverage is 1. □

Theorem 1. The thresholds $\hat{t}(\alpha)$ depend only on $\mathcal{A}$ and are fixed given $\mathcal{A}$. Conditionally on $\mathcal{A}$ and on the labels of the kept variants of $\mathcal{B}$, the class-1 kept variants of $\mathcal{B}$ are n_1 independent draws from $Q(\cdot \mid Y = 1)$, so the number of them called LoF is $\mathrm{Binomial}(n_1, \mathrm{TP}(\alpha))$ with $\mathrm{TP}(\alpha) = \Pr(\text{call LoF} \mid Y = 1)$, and likewise for the other three counts. Each one-sided Clopper–Pearson bound [4] at level $\delta/(2|G|)$ fails with probability at most $\delta/(2|G|)$; the conventions for $x = 0$, $x = n$ and $n = 0$ in Section 3 give the trivial bounds 0 and 1, which never fail. For LR^+ there are $2|G|$ such bounds ($\underline{\mathrm{TP}}$ and $\overline{\mathrm{FP}}$ for each α), so by the union bound all of them hold simultaneously with probability at least $1 - \delta$. On that event $\mathrm{TP}(\alpha) \ge \underline{\mathrm{TP}}(\alpha)$ and $\mathrm{FP}(\alpha) \le \overline{\mathrm{FP}}(\alpha)$ for every $\alpha \in G$, hence $\mathrm{LR}^+(\hat{t}(\alpha)) \ge L^+(\alpha)$ for every α (with $\mathrm{LR}^+ = \infty$ when $\overline{\mathrm{FP}}(\alpha) = 0$), including any $\hat{\alpha}^+$ chosen from these data. Since this probability bound holds for every $\mathcal{A}$, it also holds unconditionally. The argument for LR^- is identical with $\overline{\mathrm{FN}}$ and $\underline{\mathrm{TN}}$. □

Corollary 1. Every quantity bounded above is a probability conditional on the true class: $\Pr(k \in C \mid Y = k)$, the four call rates, and their ratios. Under Assumption 2 these are the same for target variants as for a test variant under Assumption 1, whatever π is. The two call sets are disjoint for any levels α^+, α^-: a LoF call at α^+ requires $q > \hat{t}_0(\alpha^+)$ and $1 - q \le \hat{t}_1(\alpha^+)$, a functional call at α^- requires $q \le \hat{t}_0(\alpha^-)$ and $1 - q > \hat{t}_1(\alpha^-)$, and both thresholds are non-increasing in α, so the first conditions contradict each other if $\alpha^- \ge \alpha^+$ and the second if $\alpha^- < \alpha^+$. A functional call on a LoF variant requires $1 \notin C$, so its rate is at most $1 - \Pr(1 \in C \mid Y = 1) \le \alpha_1$. None of these quantities involves π, so no prior is needed to certify them; this concerns the likelihood ratio with respect to the assay outcome, not evidence about pathogenicity. □

S2 Simulation

Scores were drawn per residue as $s = 2y + u_c + e$, with a residue effect $u_c \sim N(0, 0.5^2)$ shared by the five variants of a residue, unit Gaussian noise, and the class shared within a residue. A Platt calibrator was fitted on 600 variants at LoF share 0.45, conformal thresholds were set on another 600 (one variant per residue), and 3,000 test variants were drawn at LoF shares from 0.05 to 0.60, with 300 replicates per share (Figure S1). Per-class thresholds kept LoF coverage at the target at every share; pooled thresholds did not, and reweighting them to the EM estimate of the share [5] kept marginal coverage near the target while LoF coverage fell further as LoF variants became rarer. Adjusting posteriors to the EM estimate removed the calibration error caused by the shift in this simulation, where the shift is a pure change of prior, unlike in the real data (main text, Section 4.2).

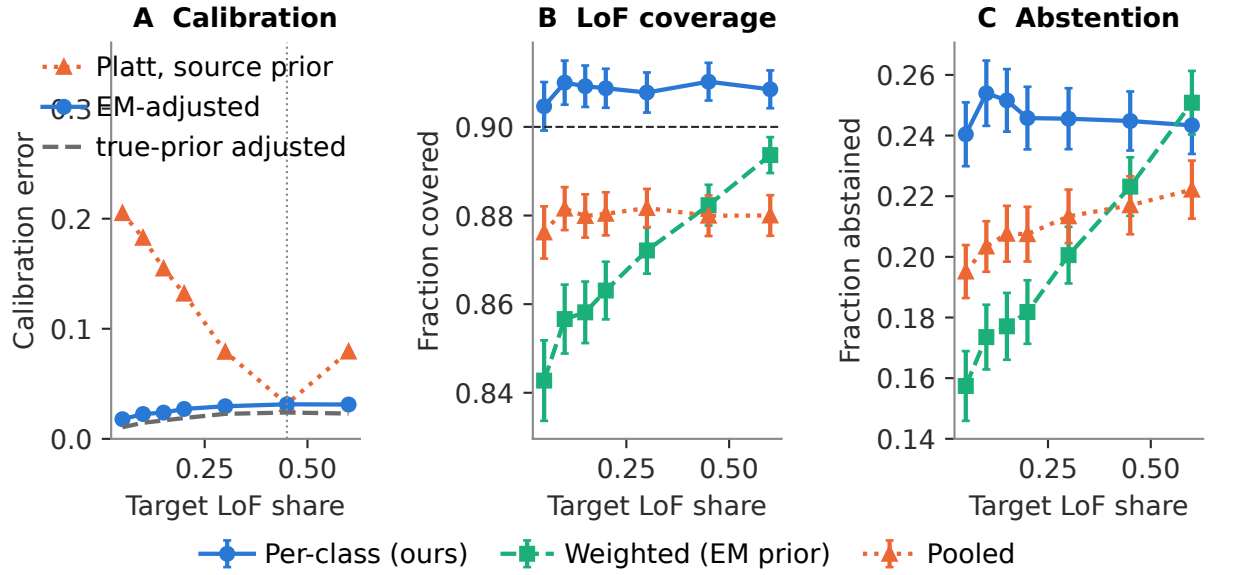


Figure S1: Simulation with residue-clustered scores, calibration at LoF share 0.45 and test share on the horizontal axis (300 replicates; error bars are 95% intervals of the mean). (A) Expected calibration error of Platt scaling at the source share, after EM prior adjustment, and after adjustment to the true share. (B) Coverage of LoF variants. (C) Abstention rate.

Why marginal thresholds do not control the minority class. A pooled split-conformal threshold $\hat{t}$ is a quantile of the mixture $\pi_c F_1 + (1 - \pi_c)F_0$ of the two class-conditional nonconformity laws, weighted by the calibration share π_c. Its coverage of class 1 is $F_1(\hat{t})$, which is not controlled; when LoF variants are rare and harder to separate, $F_1(\hat{t}) < 1 - \alpha$ while marginal coverage is still near $1 - \alpha$. Reweighting to a known test prior [5] restores marginal coverage at that prior but does not control $F_1(\hat{t})$ either.

S3 Calibration size, region and gene transfer, and per-gene results

Table S1 gives per gene and predictor the residue-weighted coverage of each class and the abstention rate of the per-class rule and two baselines; Table S2 gives the estimated LoF shares and calibration errors behind Section 4.2 of the main text. Label-shift-weighted conformal prediction [5] is omitted from the main comparison: within a gene the calibration and test halves share their LoF share, so with the true share it nearly coincides with pooled conformal, and with the EM estimate it inherits the estimation error of Section 4.2 of the main text, which collapsed its LoF coverage in some pairs.

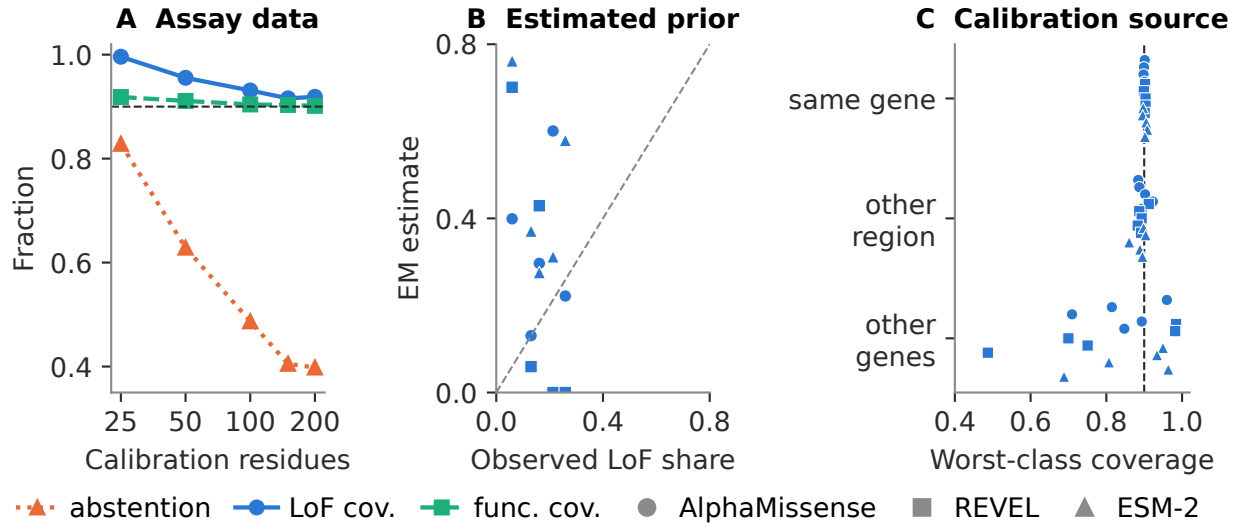


Figure S2: (A) Coverage and abstention against the number of calibration residues drawn from the gene (mean over genes, predictors and 50 draws). (B) EM estimate of the LoF share from unlabelled scores against the observed share; the dashed line is equality. (C) Worst-class coverage when calibrating on random residues of the same gene, on the other three quarters of the protein, or on the other four genes.

S4 Strata and assay-label sensitivity

For one within-gene split per gene and predictor ($\alpha = 0.1$, variant-weighted), coverage of LoF and functional variants was 0.966 and 0.929 on the third of variants farthest from the assay's class boundary, 0.929 and 0.912 on the middle third, 0.865 and 0.873 on the nearest third, and 0.881 and 0.896 on variants classified VUS in ClinVar, with abstention rates of 0.334, 0.385, 0.413 and 0.392.

Table S3 gives, for each label rule, the number of labelled variants, their LoF share, the coverage and abstention of the per-class rule, and the oracle check of the certified selection (main text, Section 4.6).

S5 Implementation

Distance from the class boundary is the absolute difference between a variant's assay score and the midpoint of the two class medians in its gene, divided by the distance between the medians. Expected calibration error uses ten equal-mass bins. AUROC intervals are percentiles of 1,000 bootstrap resamples of residues. The region split divides each gene's assayed residues into four contiguous blocks with equal numbers of distinct positions. ClinVar records whose aggregate germline classification is empty (somatic-only submissions) are treated as unclassified. All random draws use NumPy generators with fixed seeds (2027 plus a fixed offset per experiment). The experiments run on a laptop CPU with Python 3.13, NumPy 2.5, SciPy 1.18, scikit-learn 1.9 and pandas 3.1; ESM-2 scoring used fair-esm 2.0 and PyTorch 2.14 on an Apple-silicon GPU. Exact versions and data checksums are in the repository.

References

[1] Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. 2005. doi: 10.1007/b106715.

[2] Anastasios N. Angelopoulos and Stephen Bates. Conformal Prediction: A Gentle Introduction. *Foundations and Trends in Machine Learning*, 16(4):494–591, 2023. doi: 10.1561/2200000101.

[3] Vladimir Vovk. Conditional validity of inductive conformal predictors. *Machine Learning*, 92(2–3): 349–376, 2013. doi: 10.1007/s10994-013-5355-6.

Table S1: Within-gene random split, $\alpha = 0.1$, mean of 50 splits. Each cell is residue-weighted LoF coverage / functional coverage / abstention rate.

Predictor	Gene	Per-class (ours)	Pooled conformal	Fixed 0.1/0.9
AlphaMissense	BRCA1	0.92/0.90/0.25	0.95/0.89/0.32	0.97/0.93/0.44
AlphaMissense	BRCA2	0.90/0.90/0.36	0.81/0.92/0.17	0.89/0.97/0.37
AlphaMissense	BAP1	0.91/0.90/0.39	0.98/0.88/0.69	0.94/0.69/0.26
AlphaMissense	RAD51C	0.90/0.91/0.17	0.82/0.93/0.10	0.96/0.97/0.37
AlphaMissense	MSH2	0.91/0.90/0.29	0.97/0.90/0.70	0.96/0.80/0.39
REVEL	BRCA1	0.91/0.90/0.34	0.96/0.88/0.48	1.00/1.00/0.98
REVEL	BRCA2	0.91/0.90/0.42	0.84/0.91/0.24	0.95/1.00/0.68
REVEL	BAP1	0.90/0.90/0.42	0.54/1.00/0.15	0.66/1.00/0.24
REVEL	RAD51C	0.91/0.90/0.27	0.63/1.00/0.20	0.68/1.00/0.23
REVEL	MSH2	0.91/0.90/0.37	0.99/0.90/0.79	0.99/0.90/0.80
ESM-2	BRCA1	0.91/0.90/0.37	0.91/0.89/0.34	0.97/0.98/0.64
ESM-2	BRCA2	0.90/0.90/0.36	0.94/0.89/0.51	0.96/0.96/0.71
ESM-2	BAP1	0.91/0.91/0.47	0.88/0.91/0.36	0.90/0.94/0.46
ESM-2	RAD51C	0.91/0.91/0.34	0.97/0.89/0.50	0.99/0.96/0.71
ESM-2	MSH2	0.92/0.90/0.46	1.00/0.89/0.80	0.99/0.82/0.68

[4] C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934. doi: 10.1093/biomet/26.4.404.

[5] Aleksandr Podkopaev and Aaditya Ramdas. Distribution-free uncertainty quantification for classification under label shift. In *Proceedings of the Thirty-Seventh Conference on Uncertainty in Artificial Intelligence*, volume 161 of *Proceedings of Machine Learning Research*, pages 844–853, 2021.

[6] Marco Saerens, Patrice Latinne, and Christine Decaestecker. Adjusting the Outputs of a Classifier to New a Priori Probabilities: A Simple Procedure. *Neural Computation*, 14(1):21–41, 2002. doi: 10.1162/089976602753284446.

[7] Zachary Lipton, Yu-Xiang Wang, and Alexander Smola. Detecting and Correcting for Label Shift with Black Box Predictors. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 3122–3130, 2018.

Table S2: Prior estimation and calibration on assayed variants outside the ClinVar labelled set. LoF share observed and estimated from unlabeled scores by EM [6] and BBSE [7]; expected calibration error (10 equal-mass bins) of the ClinVar-Platt posterior, after EM prior adjustment, after adjustment to the true prior, and of Platt scaling fitted on the other four genes' assay labels. AM: AlphaMissense.

		LoF share			Expected calibration error			
Pred.	Gene	obs.	EM	BBSE	ClinVar Platt	+ EM	+ true prior	other genes
AM	BRCA1	0.161	0.297	0.316	0.203	0.135	0.064	0.049
AM	BRCA2	0.130	0.131	0.178	0.132	0.032	0.032	0.041
AM	BAP1	0.213	0.601	0.588	0.341	0.388	0.255	0.119
AM	RAD51C	0.259	0.222	0.252	0.067	0.043	0.037	0.162
AM	MSH2	0.060	0.399	0.395	0.361	0.340	0.120	0.139
REVEL	BRCA1	0.161	0.429	0.336	0.285	0.268	0.058	0.140
REVEL	BRCA2	0.130	0.059	0.149	0.176	0.071	0.025	0.111
REVEL	BAP1	0.213	0.000	0.000	0.092	0.213	0.149	0.138
REVEL	RAD51C	0.259	0.000	0.000	0.143	0.259	0.185	0.187
REVEL	MSH2	0.060	0.701	0.577	0.485	0.642	0.106	0.329
ESM-2	BRCA1	0.161	0.276	0.304	0.212	0.114	0.050	0.044
ESM-2	BRCA2	0.130	0.371	0.371	0.288	0.241	0.075	0.035
ESM-2	BAP1	0.213	0.312	0.327	0.171	0.121	0.086	0.092
ESM-2	RAD51C	0.259	0.579	0.538	0.246	0.320	0.122	0.102
ESM-2	MSH2	0.060	0.762	0.670	0.531	0.702	0.165	0.234

Table S3: Sensitivity to the assay-label rule. Each gene is relabelled with the authors' classes or with the control rule at the stated posterior cut-off; other genes are unchanged. Per-class: lowest mean per-class coverage over the three predictors (50 random halvings, $\alpha = 0.1$); abstention: range over predictors; bounds held: oracle selections whose certified bound the true likelihood ratio respected (100 draws per predictor); LoF-supp.: share of draws certifying LoF-supporting calls, range over predictors.

Gene	Labels	n	LoF	Per-class	Abstention	Bounds held	LoF-supp.
BRCA2	controls, 0.80	3,971	0.136	0.894	0.49–0.53	294/295	0.63–0.96
BRCA2	controls, 0.95 (main)	3,435	0.130	0.896	0.36–0.42	423/423	0.90–0.99
BRCA2	controls, 0.99	2,341	0.151	0.897	0.30–0.37	591/593	0.94–1.00
BAP1	authors (main)	4,294	0.213	0.901	0.39–0.47	460/460	0.90–1.00
BAP1	controls, 0.95	4,221	0.100	0.899	0.13–0.19	777/777	1.00
RAD51C	authors (main)	2,225	0.259	0.895	0.18–0.28	558/561	0.84–0.99
RAD51C	controls, 0.95	832	0.538	0.902	0.07–0.19	868/868	0.98–1.00